System Administration Reference Cheatsheet

Modern Linux (systemd) | Legacy Linux (SysVinit) | Windows — Service, Package, File, and Backup Management

SERVICE MANAGEMENT

Modern Linux (systemd)

`systemctl start NAME`	Start a service
`systemctl stop NAME`	Stop a service
`systemctl restart NAME`	Restart a service
`systemctl reload NAME`	Reload config (no restart)
`systemctl enable NAME`	Start at boot
`systemctl disable NAME`	Do not start at boot
`systemctl status NAME`	Status and recent logs
`systemctl is-active NAME`	Check if running
`systemctl is-enabled NAME`	Check if boot-enabled
`systemctl list-units --type=service`	All loaded services
`systemctl list-unit-files`	All installed services
`systemctl daemon-reload`	Reload after unit file edits
`systemctl mask NAME`	Prevent starting entirely
`journalctl -u NAME -f`	Follow logs for service

Legacy Linux (SysVinit)

`service NAME start|stop|restart`	Control a service
`service NAME status`	Show status
`chkconfig NAME on`	Enable at boot (RHEL 5–6)
`chkconfig NAME off`	Disable at boot
`chkconfig --list`	List all services/runlevels
`update-rc.d NAME defaults`	Enable at boot (Debian 6–7)
`update-rc.d NAME remove`	Disable at boot
`/etc/init.d/NAME start`	Direct init script control

Windows

`sc query NAME`	Show service status
`sc start NAME / sc stop NAME`	Start or stop
`sc config NAME start= auto`	Set to auto-start
`sc config NAME start= disabled`	Disable a service
`sc qc NAME`	Show config (binary path)
`net start / net stop NAME`	Start or stop
`net start`	List running services
`Get-Service NAME`	PS: service status
`Restart-Service NAME`	PS: restart
`Set-Service NAME -StartupType Auto`	PS: change startup type
`sc query type= service state= all`	List every service

CREATING / EDITING SERVICES

systemd unit file: /etc/systemd/system/NAME.service

```
[Unit]
Description=My Custom Service
After=network.target

[Service]
Type=simple
ExecStart=/usr/local/bin/myapp
Restart=on-failure
User=appuser
WorkingDirectory=/opt/myapp

[Install]
WantedBy=multi-user.target
```

Then: `systemctl daemon-reload && systemctl enable --now NAME`

SysVinit

Create `/etc/init.d/NAME` (executable shell script with LSB header). Functions: `start()`, `stop()`, `restart()`, `status()`. Then `chmod +x` and `chkconfig --add NAME` or `update-rc.d NAME defaults`.

Windows

`sc create NAME binPath= "C:\path\app.exe"`	Create a service
`sc delete NAME`	Delete a service
`sc config NAME binPath= "..."`	Change binary path
`sc failure NAME reset= 86400 actions= restart/60000`	Auto-restart on fail
`New-Service -Name NAME -BinaryPathName "path"`	PS: create service

PACKAGE MANAGEMENT

Debian / Ubuntu (apt)

`apt update`	Refresh package index
`apt install PKG`	Install a package
`apt remove PKG`	Remove (keep config)
`apt purge PKG`	Remove + delete config
`apt upgrade`	Upgrade all packages
`apt search TERM`	Search for a package
`apt list --installed`	List installed packages
`dpkg -l`	List all installed (detailed)
`dpkg -L PKG`	Files owned by package
`dpkg -S /path/to/file`	What package owns this file?

RHEL / CentOS / Fedora (dnf/yum)

`dnf install PKG`	Install (or yum on RHEL 5–7)
`dnf remove PKG`	Remove a package
`dnf update`	Update all packages
`dnf search TERM`	Search for a package
`dnf list installed`	List installed packages
`rpm -qa`	List all installed (rpm level)
`rpm -ql PKG`	Files owned by package
`rpm -qf /path/to/file`	What package owns this file?

Other Distros

`zypper install PKG / zypper update`	openSUSE / SLES
`pacman -S PKG / pacman -Syu`	Arch Linux
`apk add PKG / apk upgrade`	Alpine Linux

Windows

`winget install PKG`	Install from winget
`winget search TERM`	Search packages
`winget upgrade --all`	Update all
`winget list`	List installed software
`choco install PKG`	Chocolatey: install
`choco upgrade all`	Chocolatey: update all
`Get-WindowsFeature`	List server roles/features
`Install-WindowsFeature NAME`	Install a role
`DISM /Online /Enable-Feature /FeatureName:NAME`	Enable Windows feature

FINDING FILES: DATABASES

Linux – MySQL / MariaDB

`find / -name "*.frm" 2>/dev/null`	Table definition files
`find / -name "ibdata*" 2>/dev/null`	InnoDB data files
`find / -name "my.cnf" -o -name "my.ini"`	Config files
`mysql -e "SHOW VARIABLES LIKE 'datadir';"`	Data dir from MySQL
`ss -tlnp | grep 3306`	Is MySQL listening?

Default configs: `/etc/mysql/my.cnf` (Debian), `/etc/my.cnf` (RHEL)

Linux – PostgreSQL

`find / -name "postgresql.conf" 2>/dev/null`	Config file
`find / -name "pg_hba.conf" 2>/dev/null`	Auth config
`su - postgres -c "psql -c 'SHOW data_directory;'"`	Data directory
`ss -tlnp | grep 5432`	Is Postgres listening?

Linux – SQLite

`find / \( -name "*.db" -o -name "*.sqlite" -o -name "*.sqlite3" \) 2>/dev/null`	All SQLite databases

Windows – SQL Server

`Get-Service MSSQL*`	Find SQL Server services
`sqlcmd -S . -Q "SELECT name, physical_name FROM sys.master_files"`	All DB file locations
`dir /s *.mdf`	Find data files
`dir /s *.ldf`	Find log files

Windows – MySQL

`sc query MySQL*`	Find MySQL services
`dir /s my.ini`	Find config file

FINDING FILES: WEB SERVERS

Linux – Apache

`apachectl -S` (or `httpd -S`)	Virtual hosts and doc roots
`apache2ctl -M`	Loaded modules
`grep -r "DocumentRoot" /etc/apache2/`	Doc roots (Debian)
`grep -r "DocumentRoot" /etc/httpd/`	Doc roots (RHEL)
`find /etc -name "*.conf" -path "*/apache*"`	All Apache configs

Linux – Nginx

`nginx -T`	Dump full parsed config
`grep -r "root " /etc/nginx/`	All document roots
`grep -r "server_name" /etc/nginx/`	All server names

Linux – Generic

`ss -tlnp | grep -E ':80 |:443 |:8080 '`	What listens on web ports?
`find /var/www -type f -name "*.php"`	PHP files in web root

Windows – IIS

`appcmd list site`	List all IIS sites
`appcmd list vdir`	Virtual directories
`Get-IISSite`	PS: list sites
`Get-WebBinding`	PS: all bindings

Default root: `C:\inetpub\wwwroot`
Config: `C:\Windows\System32\inetsrv\config\applicationHost.config`
appcmd lives at `%windir%\system32\inetsrv\appcmd.exe`

FINDING FILES: OTHER SERVICES

Linux

`ss -tlnp`	All listening TCP + process
`ss -ulnp`	All listening UDP + process
`find /etc -name "named.conf*"`	BIND DNS config
`grep -r "zone " /etc/named* /etc/bind/*`	DNS zone files
`postconf -n`	Postfix non-default settings
`cat /etc/vsftpd.conf`	vsftpd FTP config
`testparm -s`	Effective Samba config
`smbstatus`	Samba connections/shares
`find /etc -name "slapd*"`	OpenLDAP config

Windows

`netstat -abno`	All ports + PIDs + names
`Get-NetTCPConnection -State Listen`	PS: listening ports
`dnscmd /enumzones`	List DNS zones
`Get-DnsServerZone`	PS: list DNS zones
`dcdiag /v`	DC diagnostics
`repadmin /replsummary`	Replication health
`nltest /dclist:DOMAIN`	List all DCs
`netsh dhcp server show scope`	DHCP scopes
`net share`	List file shares
`Get-SmbShare`	PS: shares

CREATING BACKUPS

Linux – Files and Directories

`tar czf /backup/etc.tar.gz /etc`	Compress /etc
`tar czf /backup/home.tar.gz /home`	Compress /home
`tar czf /backup/www.tar.gz /var/www`	Compress web roots
`tar xzf /backup/etc.tar.gz -C /restore`	Extract a backup
`rsync -av /src/ /dst/`	Incremental sync
`rsync -av -e ssh /src/ user@host:/dst/`	Remote sync via SSH
`cp -a /etc /backup/etc-$(date +%F)`	Copy with timestamp

Linux – Database Backups

`mysqldump -u root -p --all-databases > all.sql`	Dump all MySQL DBs
`mysqldump -u root -p DBNAME > db.sql`	Dump one MySQL DB
`mysql -u root -p DBNAME < db.sql`	Restore MySQL
`pg_dumpall -U postgres > all.sql`	Dump all PostgreSQL
`pg_dump -U postgres DBNAME > db.sql`	Dump one PostgreSQL DB
`psql -U postgres DBNAME < db.sql`	Restore PostgreSQL
`sqlite3 /path/db .dump > bak.sql`	Dump SQLite

Linux – Quick Backup Script

```
#!/bin/bash
D=$(date +%F-%H%M); B=/backup/$D; mkdir -p $B
tar czf $B/etc.tar.gz /etc
tar czf $B/www.tar.gz /var/www /srv 2>/dev/null
mysqldump -u root --all-databases | gzip > $B/mysql.sql.gz
cp /etc/fstab /etc/passwd /etc/shadow /etc/group $B/
crontab -l > $B/root-crontab.txt 2>/dev/null
systemctl list-unit-files > $B/services.txt
echo "Backup complete:  $B"
```

Windows – System and Files

`wbadmin start systemstatebackup -backupTarget:E:`	System State backup
`wbadmin start backup -backupTarget:E: -include:C:`	Full volume backup
`robocopy C:\src E:\dst /MIR /ZB /R:1 /W:1`	Mirror copy

Windows – Registry, Policy, Firewall

`reg export HKLM\SOFTWARE out.reg`	Export registry key
`reg import out.reg`	Import registry key
`secedit /export /cfg sec.cfg`	Export security policy
`netsh advfirewall export fw.wfw`	Export firewall rules
`netsh advfirewall import fw.wfw`	Import firewall rules

Windows – AD and Database

`ntdsutil "ac i ntds" "ifm" "create full C:\bak" q q`	AD IFM backup
`Backup-GPO -All -Path C:\GPOBackup`	Backup all GPOs
`sqlcmd -Q "BACKUP DATABASE [db] TO DISK='C:\b.bak'"`	SQL Server backup

QUICK TRIAGE: WHAT CHANGED?

Linux

`find / -mmin -60 -type f 2>/dev/null`	Files modified in last hour
`find / -mtime -1 -type f 2>/dev/null`	Files modified in last 24h
`last -20`	Last 20 logins
`lastb -20`	Last 20 failed logins
`history`	Command history (current user)
`cat /home/*/.bash_history`	All users' histories
`ausearch -ts recent`	Recent audit events
`debsums -c`	Changed package files (Debian)
`rpm -Va`	Changed package files (RHEL)

Windows

`Get-ChildItem C:\ -Recurse | Where {$_.LastWriteTime -gt (Get-Date).AddHours(-1)}`	Files modified in last hour
`wevtutil qe Security /c:20 /f:text`	Last 20 security events
`Get-EventLog Security -Newest 50`	Recent security events
`query user`	Who is logged in?
`qwinsta`	All sessions (incl. RDP)
`sfc /verifyonly`	System file integrity

USER AND PERMISSION MANAGEMENT

Linux

`useradd -m USERNAME`	Create user with home dir
`userdel -r USERNAME`	Delete user + home dir
`passwd USERNAME`	Change password
`usermod -aG GROUP USER`	Add user to group
`gpasswd -d USER GROUP`	Remove from group
`chown USER:GROUP FILE`	Change ownership
`chmod 750 FILE`	Set permissions (rwxr-x—)
`grep "sudo|wheel" /etc/group`	Who has sudo?
`visudo`	Edit sudoers safely
`passwd -l USERNAME`	Lock account
`passwd -u USERNAME`	Unlock account
`chage -l USERNAME`	Password expiry info

Windows

`net user USERNAME PASS /add`	Create local user
`net user USERNAME /del`	Delete user
`net localgroup GROUP USER /add`	Add to group
`net localgroup GROUP USER /del`	Remove from group
`net localgroup Administrators`	List local admins
`icacls FILE`	View permissions
`icacls FILE /grant USER:F`	Grant full control
`takeown /F FILE /R`	Take ownership
`net user USERNAME /active:no`	Disable account

Windows – Active Directory

`New-ADUser -Name "user" -Enabled $true`	Create AD user
`Add-ADGroupMember "Group" "User"`	Add to AD group
`Disable-ADAccount -Identity USER`	Disable AD account
`Search-ADAccount -LockedOut`	Find locked accounts
`Unlock-ADAccount -Identity USER`	Unlock account

FIREWALL QUICK REFERENCE

Linux – iptables

`iptables -L -n -v`	List all rules
`iptables -A INPUT -p tcp --dport 22 -j ACCEPT`	Allow SSH
`iptables -A INPUT -s IP -j DROP`	Block an IP
`iptables -P INPUT DROP`	Default deny inbound
`iptables-save > /etc/iptables.rules`	Save rules
`iptables-restore < /etc/iptables.rules`	Restore rules

Linux – firewalld (RHEL 7+)

`firewall-cmd --list-all`	Show current rules
`firewall-cmd --add-port=80/tcp --permanent`	Open port
`firewall-cmd --reload`	Apply changes

Windows

`netsh advfirewall show allprofiles`	Firewall status
`netsh advfirewall set allprofiles state on`	Enable firewall
`netsh advfirewall firewall add rule name="Allow SSH" dir=in action=allow protocol=TCP localport=22`	Allow a port inbound
`netsh advfirewall firewall add rule name="Block" dir=in action=block remoteip=IP`	Block an IP
`netsh advfirewall firewall show rule name=all`	List all rules

COMMON CONFIG FILE LOCATIONS

Linux

`/etc/ssh/sshd_config`	SSH server config
`/etc/passwd, /etc/shadow`	User accounts / passwords
`/etc/group, /etc/gshadow`	Groups
`/etc/sudoers, /etc/sudoers.d/`	Sudo rules
`/etc/fstab`	Filesystem mounts
`/etc/hosts, /etc/resolv.conf`	DNS / hostname resolution
`/etc/crontab, /var/spool/cron/`	Scheduled tasks
`/etc/rsyslog.conf`	Syslog config
`/var/log/auth.log` (Deb)	Authentication log
`/var/log/secure` (RHEL)	Authentication log
`/var/log/syslog` (Deb)	System log
`/var/log/messages` (RHEL)	System log

Windows

`C:\Windows\System32\drivers\etc\hosts`	Hosts file
`C:\Windows\System32\config\`	SAM, SECURITY, SYSTEM
`C:\Windows\System32\winevt\Logs\`	Event log files
`C:\inetpub\wwwroot\`	IIS default web root
`C:\ProgramData\`	App config/data (hidden)
`%APPDATA%`	Per-user app config